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Structure-property-performance relationships of lactic acid-based deep eutectic solvents with different hydrogen bond acceptors for corn stover pretreatment

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HIGHLIGHTS

- Ten types of lactic acid-based DES by incorporating differently structured HBAs.
- HBAs with short alkyl chain achieved better DES pretreatment results.
- The functional groups of HBAs significantly affected DES pretreatment efficacy.
- Hydrogen bond acidity of DES positively correlated with pretreatment performance.
- Structure-function-performance relationship of DES pretreatment was clarified.

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ABSTRACT

Herein, ten types of lactic acid-based deep eutectic solvents (DESs) with differently structured hydrogen bond acceptors (HBAs) were used for corn stover pretreatment. Among the tested DESs, those composed of HBAs with short alkyl chain were more effective to remove lignin and xylan, resulting in higher enzymatic digestion of the pretreated solids than their counterparts with long alky chain. Also, functional groups of HBAs demonstrated significant effects on biomass deconstruction. In order to interpret the different pretreatment performance of the tested DESs, Kamlet-Taft solvent polarity parameters of the tested DESs were correlated to their lignocellulose pretreatment performance. It was found that hydrogen bond acidity (Kamlet-Taft α parameter) had strong positive relationships with pretreatment efficacy of the studied DESs. These findings not only clarified the structure–property-performance relationships of the DESs, but also provided novel insights into design and selection of DESs for lignocellulose pretreatment.

1. Introduction

Lignocellulosic biomass, composed of cellulose, hemicellulose and lignin, is currently the largest available renewable resource with a great potential as feedstock for large-scale production of a wide range of value-added products including transport fuels, platform chemicals and functional materials. To achieve the low-cost conversion of biomass into these products, it is widely accepted that the key lies in efficient depolymerization of polysaccharide fraction (hemicellulose and cellulose) of raw biomass into oligo- and monosaccharides, which are readily fermented by microorganism to produce useful products (Aristizábal-Marulanda & Cardona Alzate, 2019). However, due to natural recalcitrance of biomass resulting from its complex structure and interactions between individual components of biomass, cost-efficient enzymatic hydrolysis of polysaccharide fraction for production of fermentable sugars is still one of the main bottlenecks that hinder the industrialization and commercialization of lignocellulosic biorefinery (Chandel et al., 2018). Therefore, development of biomass pretreatment technologies to reduce its recalcitrance has been the research hotspot in the field of biomass bioconversion in the past decades (Tian et al., 2018).

In recent years, in the context of green chemistry, researchers around the world have been devoting efforts to developing green processes for

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pretreating biomass under mild conditions, which can maximize full utilization of biomass, minimize waste production, and avoid the use of toxic and hazardous chemicals (Hassan et al., 2018). Among these developed green pretreatment processes, deep eutectic solvents (DESs) stand out as important solvents for biomass pretreatment due to their unique properties like low-cost of starting chemicals, easy synthesis without purification step and waste generation, low/no toxicity, high biodegradability, etc. (Hansen et al., 2021; Smith et al., 2014). DESs are simply prepared by mixing hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD) at certain molar ratios under heating conditions, in which HBA and HBD can self-associate to form homogenous and eutectic mixture through hydrogen bond interactions with melting point significantly lower than each of its components (Zhang et al., 2012). DES pretreatment of biomass has attracted the extensive attention of scientific and technologic community, which can be reflected in the published reviews on this topic in recent years (Satlewal et al., 2018; Tan et al., 2020; Wang et al., 2021; Xu et al., 2020). It can be seen from analysis of available reports that more attentions are paid to investigating the effect of HBDs on DES pretreatment of lignocellulose, and it was found that structure of HBDs containing carboxyl, hydroxyl or amine/amide group(s) was highly correlated to delignification capability of DES (Hong et al., 2020b). On the other hand, little attention was paid to exploring the influence of HBAs on biomass deconstruction with DES. HBAs generally includes various quaternary ammonium, sulfonium and phosphonium salts (Zhang et al., 2012). Ling et al. (2020) pretreated bamboo with levulinic acid (LevA)-based DESs with acetamide (Am), betaine (Ba) and choline chloride (ChCl) as HBAs. They found that ChCl-LevA DES pretreatment demonstrated the best result with 79.07% cellulose saccharification rate of pretreated samples. Guo et al. (2019) carried out corncob pretreatments with benzyltrimethylammonium chloride (BTMAC)-lactic acid (LA) and benzyltriethylammonium chloride (BTEAC)-LA and reported that better pretreatment results were obtained from BTMAC-LA pretreatment, which removed 80.8% xylan and 63.4% lignin. With those few studies, it is difficult to draw reliable conclusions about the effect of HBAs with different structures on biomass deconstruction using DES. More in-depth investigations are, therefore, needed to be conducted to elucidate the relationship between HBA structure and biomass pretreatment performance of DESs.

To understand the rules governing the biomass fractionation process during DES pretreatments, it is important to determine the polarity of DES, which characterizes solvent strength and refers to interactions between DES solvent and biomass substrate. Polarity of DES is sensitive towards the structures of both HBA and HBD, which associate together to form DES. Solvatochromic probes have been used to characterize the polarity of a solvent, in particular, the quantitative Kamlet-Taft parameters were broadly used in literature to quantify a solvent's hydrogen bond acidity (parameter α), the hydrogen bond basicity (parameter β), and dipolarity/polarizability (parameter π*). Kamlet-Taft parameters of a DES are reported to be closely related to its performance in biomass pretreatment. Oh et al. (2020) performed pine wood pretreatment with different types of ChCl-based DES and found that the hydrogen bond acidity (parameter α) of the DESs demonstrated a positive linear relationship to lignin extraction. Liu et al. (2019) and Hong et al. (2020a) investigated the effect of different DESs on dissolution of separated lignin and concluded that lignin-dissolving ability was correlated to the difference between the values of α and β (α-β) of DESs. Based on the existing data, Hong et al. (2020b) correlated Kamlet-Taft parameters with lignin extraction ability and concluded that α and β values of DESs are more strongly positively related to delignification than the π* value. It was reported that the β parameter could be used to predict lignocellulose pretreatment efficacy of ionic liquid perfectly (Brandt et al., 2013). Clarifying the relationship between Kamlet-Taft parameters of DESs and measurable results of lignocellulosic biomass treated with these solvents would greatly improve the design of DES and simplify current time-consuming selection processes of DES used for lignocellulose pretreatment.

In this present work, a series of LA-based DES with differently structured HBAs were synthesized and applied to pretreat corn stover (CS) for investigating the effect of HBA structures (alkyl chain length and functional groups) on effectiveness of DES pretreatment, which was evaluated by removal of lignin and xylan, enzymatic digestibility of pretreated solid and recovery of extracted lignin. Most importantly, solvatochromic parameters of the synthesized DESs, including polarity dipolarity/polarizability (parameter π*), hydrogen bond acidity (parameter α) and hydrogen bond basicity (parameter β) were determined using a set of dye probes and correlated to the pretreatment efficacy of the tested DESs, with an attempt to establish structure–property-performance relationship of DESs during biomass processing and elucidate the pretreatment mechanism of DESs with different structures.

2. Materials and methods

2.1. Materials

CS was obtained from a farm in Hengshui, Hebei, China. It was rinsed with water to remove impurities, dried in an oven at 60 °C to constant weight and then milled and sieved to obtain particles with size ranging from 60 to 80 mesh for pretreatment experiments. The chemical compositions of CS based on the dry mass of raw biomass were as follows: 32.39% glucan, 22.94% xylan, 21.91% lignin (20.33% acid-insoluble lignin and 1.58% acid-soluble lignin) and 4.61% ash. Cellulase, β-glucosidase and xylanase were all purchased from Sunson Industrial Group Co., Ltd., Ningxia, China. The filter paper activity (FPA) of the cellulase was 112 filter paper unit (FPU)/mL, the activities of the β-glucosidase and the xylanase were 118 cellobiose unit (CBU)/mL and 1369 xylan unit (XU)/mL, respectively.

LA, ChCl, tetramethyl ammonium chloride (TMAC), tetraethyl ammonium chloride (TEAC), tetrapropyl ammonium chloride (TPAC), tetrabutyl ammonium chloride (TBAC) and methyl triethyl ammonium chloride (MTAC) were acquired from Shanghai Aladdin Biochemical Technology Co., Ltd., China. BTMAC, BTEAC and chlorocholine chloride (CCC) were purchased from Beijing MREDA Technology Co., Ltd, China. Allyl trimethyl ammonium chloride (ATMAC) was obtained from Beijing OKA Biotechnology Co., Ltd. All above chemical reagents have purity of no less than 98%. 4-nitroaniline and 4-nitroanisole with purity ≥99% were obtained from Beijing J&K Scientific, Ltd. Nile Red with purity ≥95% were purchased from Shanghai Acmec Biochemical Co., Ltd, China.

2.2. DES synthesis

Different DES solutions were synthesized by mixing different HBAs with LA at a molar ratio of 1:2 and then heated in an oil bath at 80 ℃ with magnetic stirring until a homogeneous and transparent liquid was obtained. The synthesized DES solutions were dried in a vacuum oven at 60 ℃ for 24 h to remove moisture and then stored in a dryer for subsequent pretreatment experiments.

2.3. DES pretreatments

CS pretreatments were conducted in 120 mL Ace pressure tubes sealed with polytetrafluoroethylene (PTFE) plugs (Ace Glass, NJ, USA). The pretreatment temperature and time were chosen according to the report of Xu et al. (2016). A slurry of 10 wt% biomass was prepared by combining 1.00 g CS with 10.00 g DES in the tube placed in a 130 °C oil bath. After 2 h of reaction, the pretreatment slurry was filtered under vacuum with a sand-core funnel (G3). The obtained solid residue was washed three times with 20 mL 50 vol% of ethanol aqueous solution and then dried at 90 °C for 24 h for further use. The filtrate from pretreatment slurry filtration and solid residue washing were combined followed by rotary-evaporation at 40 °C to remove ethanol. The precipitated

lignin, referred to DES lignin thereafter, after ethanol removal was centrifuged at 10,000 g for 5 min and then dried at 40 °C under vacuum for further analysis.

After CS pretreatment, solid recovery and component (glucan, xylan or lignin) recovery were defined in the following equations:

$$Solid\ recovery = (mass\ of\ solid\ recovered\ after\ pretreatment) \times 100\% / (mass\ of\ raw\ biomass) \quad (1)$$

$$Component\ (glucan,\ xylan\ or\ lignin)\ recovery = (percent\ content\ of\ certain\ component\ in\ pretreated\ biomass) \times (solid\ recovery) \times 100\% / (percent\ content\ of\ corresponding\ component\ in\ raw\ biomass) \quad (2)$$

Component (glucan, xylan or lignin) removal was calculated using below equation:

$$Component\ (glucan,\ xylan\ or\ lignin)\ removal = 100\% - corresponding\ component\ recovery\ (\%) \quad (3)$$

DES lignin yield was calculated according to the following equation:

$$DES\ Lignin\ yield = (mass\ of\ DES\ lignin) \times 100\% / (mass\ of\ lignin\ in\ raw\ biomass) \quad (4)$$

2.4. Enzymatic hydrolysis

Enzymatic hydrolysis of pretreated CS was carried out in citrate buffer (0.05 M, pH 4.8) in 50 mL conical flask at 2% (w/v) solid concentration. Cellulase, β-glucosidase and xylanase were added to the flask at enzyme loadings of 50 FPU, 100 CBU and 500 XU per g dry substrate, respectively. In addition, Sodium azide (0.02%) was also added to inhibit the growth of microorganisms. Enzymatic hydrolysis was carried out in a rotary incubator at 50 °C and 150 rpm. Samples were withdrawn at different time intervals, filtered through 0.22 μm nylon syringe filter and analyzed by high-performance liquid chromatography (HPLC).

Glucose yield from enzymatic digestibility of pretreated CS was calculated using following equation:

$$Glucose\ yield = (glucose\ content\ in\ hydrolysate) \times 100\% / (glucan\ content\ in\ substrate \times 1.11) \quad (5)$$

2.5. Analytical methods

2.5.1. Characterization of DESs

The viscosity and density of DES was measured using DMA5000M-Lovis2000ME densitometer/viscometer (Anton Paar, Graz, Austria). The conductivity and pH of DES were determined by a DDS-307A conductivity meter (Shanghai Rex Instrument, China) and a SevenCompact S220 pH meter (Mettler Toledo, Switzerland), respectively.

As for measurement of Kamlet-Taft polarity parameters of DES, 4-nitroanisole was used as a dye indicator for determining the value of π* (Laurence et al., 1994), which reflects the dipolarity/polarizability of solvent. 4-nitroaniline and Nile red were respectively used as solvatochromic probes to calculate the values of β and α (Liu et al., 2019), which measures hydrogen bond alkalinity and hydrogen bond acidity, respectively. The concentrations of dyes in methanol solution used for determination were 0.05 mM for both 4-nitroanisole and 4-nitroaniline and 1 mM for Nile red. The dye solution (2 mL) was first transferred into a centrifuge tube and then dried at 40 °C under vacuum for 6 h to remove methanol. Next, 1.00 g DES was added into the tube and sonicated at 25 °C to dissolve the dye. The adsorption spectra of the dye-DES mixture were recorded on a Mettler Toledo UV5Bio UV–vis spectrophotometer (Columbus, OH, USA).

The solvatochromic parameters were calculated using the following equations (Jessop et al., 2012; Liu et al., 2019):

$$\pi^* = 14.57 - 4270/\lambda_{max,OMe} \quad (6)$$

$$\beta = 11.134 - 3580/\lambda_{max,NH_2} - 1.125 \times \pi^* \quad (7)$$

$$\alpha = (19.9657 - 1.0241 \times \pi^* - \nu_{NR})/1.6078 \quad (8)$$

$$\nu_{NR} = 1/(\lambda_{max,NR} \times 10^{-4}) \quad (9)$$

where $\lambda_{max,OMe}$, λ_{max,NH_2} and $\lambda_{max,NR}$ represent the maximum absorption wavelength of 4-nitroanisole, 4-nitroaniline and Nile Red, respectively.

2.5.2. Compositional analysis of biomass composition, DES lignin and pretreatment liquids

The chemical compositions of CS samples before and after pretreatment were quantified according to the recommended US National Renewable Energy Laboratory (NREL) protocol (Sluiter et al., 2008). The polysaccharide content associated with DES lignin was determined by one-step dilute H_2SO_4 hydrolysis as described by Guo et al. (2019). The contents of sugars (xylose and glucose) and pretreatment inhibitors in the pretreatment liquids, i.e., acetic acid, furfural and 5-hydroxymethylfurfural (HMF), were determined using HPLC following the methods described by Li et al. (2020).

2.5.3. Enzyme activity determination

FPA of cellulase and β-glucosidase activity were measured based on the procedures described by Ghose (1987) using Whatman No.1 filter paper and cellobiose as substrates, respectively. Xylanase activity was determined according to the method described by Bailey et al. (1992) with oat spelt xylan as substrate.

3. Results and discussion

3.1. Synthesis and characterization of DESs

In the present work, ten types of HBA with different chemical structures were mixed with LA HBD at 1:2 mol ratio and heated to prepare DESs. All the individual components dissolved well, and finally obtaining colorless and homogeneous DES solutions. The synthesized DESs were very stable and there were no crystals even after three months of storage in sealed bottles. These DES solutions were subsequently characterized in terms of density, viscosity, ionic conductivity, pH and polarity at room temperatures.

The polarity characteristics of a solvent is a key parameter reflecting its overall solvation capability towards a solute. It has been reported to affect reaction rate, product yield and selectivity as well as reaction mechanism during a chemical reaction process (Katritzky et al., 2004). Although DESs hold great promise as green and sustainable alternative solvents to commonly used environmentally harmful organic solvents, few data on the polarity of DESs has been reported previously (Kundu et al., 2020). The polarity property of a solvent is often estimated by the analysis of shifting UV–vis absorbance spectra of solvatochromic probes (dyes). The most frequently used method to describe the polarity property of a solvents is the multiparametric Kamlet-Taft polarity scale (Weiß et al., 2021), which quantifies hydrogen bond acidity (α), hydrogen bond basicity (β) and dipolarity/polarizability (π*) of a solvent. These parameters together would comprehensively describe a solvent's polarity. it is worthwhile mentioning that the polarity scales are probe dependent, which means that we cannot compare polarity parameters obtained by different solvatochromic probes.

As aforementioned, polarity properties of DES are reported to be linked to their biomass pretreatment performance in terms of delignification and hemicellulose removal. The measured Kamlet–Taft solvent polarity parameters of the synthesized DESs are presented in Table 1. Three solvatochromic parameters, i.e., π*, β and α, were used to indicate the polarizability, basicity, and acidity of the DES solvents, respectively.

Table 1
The Kamlet-Taft parameter values for the tested DES solutions.

No.	DES	Kamlet − Taft parameters		
		π*	β	α
1	TMAC-LA	1.117	0.516	0.990
2	MTAC-LA	1.111	0.525	0.944
3	TEAC-LA	1.109	0.554	0.800
4	TPAC-LA	1.104	0.579	0.675
5	TBAC-LA	1.083	0.617	0.614
6	BTMAC-LA	1.117	0.574	0.953
7	BTEAC-LA	1.092	0.593	0.750
8	ChCl-LA	1.125	0.511	0.970
9	CCC-LA	1.066	0.548	0.793
10	ATMAC-LA	0.867	0.692	0.779

The parameter π* measures the ability of the solvent to stabilize a charge or a dipole by virtue of its dielectric effect (Kamlet et al., 1977). The π* values of investigated DESs ranged from 0.867 to 1.125. Higher π* values indicate the studied DESs are polarizable solvents. ATMAC-LA demonstrated the lowest π* value, while the highest π* value was attained from ChCl-LA. The π* values of synthesized LA-based DESs decreased in the following HBA order: ChCl > TMAC = BTMAC > MTAC > TEAC > TPAC > BTEAC > TBAC > CCC > ATMAC. By examining the HBA structures, it could be found that HBAs with short alkyl chain result in DESs with high π* values compared to that with long alkyl chain (TMAC vs MTAC vs TEAC vs TPAC vs TBAC, BTMAC vs BTEAC). In addition, the functional groups of HBA also influence π* values of DESs. HBAs that contains hydroxyl (–OH, ChCl), chloro (-Cl, CCC) and carbon–carbon double bond (-C = C, ATMAC) groups lead to DESs with π* values in the following order: –OH > -Cl > -C = C.

The parameter β represents a solvent's ability to donate electron density to form a hydrogen bond with a solute's protons. The β values of tested DESs varied between 0.511 and 0.692, with the former value (the lowest level) being obtained from ChCl-LA DES and the latter (the highest level) from ATMAC-LA DES. The β values of prepared LA-based DESs decreased in the following HBA order: ATMAC > TBAC > BTEAC > TPAC > BTMAC > TEAC > CCC > MTAC > TMAC > ChCl. By comparison, it was evident that the effect of alkyl chain length (TMAC vs MTAC vs TEAC vs TPAC vs TBAC and BTMAC vs BTEAC) and functional group (ChCl vs CCC vs ATMAC) of HBA on β values of DES was opposite to that on their π* values.

The parameter α describes the ability of a solvent to donate a proton in a solvent-to-solute hydrogen bond. The α values of the examined DESs values fluctuated between 0.614 and 0.990. TBAC-LA exhibited the lowest α value, whereas TMAC-LA demonstrated the highest. As indicated by these high α values, all the tested DESs possesses relatively high hydrogen bond donating ability. The high acidic nature of the DESs is likely due to the carboxylic functional group present in the DESs. The α values of the DESs decreased in the following HBA order: TMAC > ChCl > BTMAC > MTAC > TEAC > CCC > ATMAC > BTEAC > TPAC > TBAC. Obviously, the effect of chemical structure of HBA on α values of DES had the same trend as that on their π* values for the categorized three groups of HBA. Higher α and lower β parameter values of these DESs classify them as hydrogen bond acidic solvents.

3.2. Pretreatment of biomass with the DESs

Ten types of HBA with different alkyl chain lengths and functional groups were used to synthesize LA-based DES, which was subsequently employed to pretreat CS. The solid recovery and composition change of CS after DES pretreatments are shown in Table 2. As for pretreatment with DES prepared by HBAs with different alkyl chain length (entries 1–7, Table 2), it was found that glucan recovery after pretreatment varied between 83.40% and 87.81%. No correlation was observed between glucan recovery and alkyl chain length of HBA used to synthesize LA-based DES. However, the removal of lignin and xylan were found to have obvious correlation with chemical structures of the HBAs. The removal of xylan and lignin deceased after pretreatment using DES composed of HBA with increasing alkyl chain length. For example, pretreatment with DES consisting of TMAC (short alkyl chain) removed 78.66% xylan and 62.57% lignin, while those two values decreased to 60.64% and 41.49%, respectively, for pretreatment using DES with TBAC (long alkyl chain) as HBA. In tandem with decreases in xylan removal and delignification, the solid recovery increased from 50.41% for TMAC-LA pretreatment to 58.48% for TBAC-LA pretreatment. This was also the case for DESs with HBA containing benzene ring. In comparison to BTMAC (short alkyl chain)-LA pretreatment, reductions in both xylan removal and delignification were observed for BTEAC (long alkyl chain)-LA pretreatment. Guo et al. (2019) also reported the better performance of BTMAC-LA than BTEAC-LA when pretreating corncobs. They attributed the superior capability of BTMAC-LA DES in terms of delignification and xylan removal to the less steric hindrance of BTMAC than BTEAC during pretreatment. It was obvious that this steric hindrance theory could also be applied to the tested DESs containing HBAs with different alkyl chain length but not containing benzene ring (Entries 1–5, Table 2).

Table 2
Solid recovery and composition change of pretreated biomass with different DESs.

No.	Pretreatment	Solid recovery (%)	Recovery (%)	Removal (%)	
			Glucan	Xylan	Lignin
1	TMAC-LA	50.41	84.99 ± 1.71	78.66 ± 0.44	62.57 ± 1.56
2	MTAC-LA	50.53	85.99 ± 1.87	76.21 ± 0.55	60.61 ± 0.67
3	TEAC-LA	52.64	87.81 ± 0.75	73.13 ± 0.53	57.72 ± 1.01
4	TPAC-LA	54.49	84.75 ± 1.60	67.39 ± 0.59	47.33 ± 1.44
5	TBAC-LA	58.48	83.40 ± 2.40	60.64 ± 0.28	41.49 ± 1.47
6	BTMAC-LA	48.39	83.69 ± 0.61	73.91 ± 0.49	63.34 ± 0.44
7	BTEAC-LA	50.13	87.35 ± 0.43	70.87 ± 0.72	60.55 ± 0.89
8	ChCl-LA	48.83	82.70 ± 0.14	77.61 ± 0.13	61.93 ± 1.49
9	CCC-LA	52.09	85.68 ± 2.09	78.02 ± 0.43	49.79 ± 1.71
10	ATMAC-LA	52.68	85.82 ± 1.95	70.24 ± 0.34	54.78 ± 1.15

Pretreatments were performed at 130 °C for 2 h at 10 wt% biomass concentration.

When it came to DES pretreatment containing HBA with different functional groups, glucan retention after pretreatment using DES with hydrolxyl HBA (ChCl) was 82.70%, slightly lower than that obtained from pretreatment using its counterparts with HBA containing chloro (ATMAC) or carbon–carbon double bond (CCC) groups, both preserving around 86% glucan. In sharp contrast to the similar effect of HBA alkyl chain length on removal of xylan and lignin, the effect of HBA functional groups on xylan removal was different from that on delignification. Pretreatment with ChCl-LA consisting of HBA with hydroxyl group removed nearly the same amount of xylan (about 78%) as that with CCC-LA composed of HBA containing chloro group, whereas the xylan removal for ATMAC-LA with HBA having carbon–carbon double bond was 70.24%. This suggested that substitution of hydroxyl or chloro groups with carbon–carbon double bond in HBA decreased the capability of DES to depolymerize xylan. As far as lignin removal was concerned, ChCl-LA DES pretreatment again demonstrated superior delignification performance (61.93%), followed by ATMAC-LA, which showed a lignin removal of 54.78%. A relatively poor capability of delignification (49.79%) was observed from CCC-LA DES pretreatment.

Therefore, the functional group of HBA on delignification after DES pretreatment followed the decreasing order: –OH > -C = C > -Cl. Liu et al. (2019) investigated the dissolution of four kinds of lignin in ATMAC- and ChCl-based DES solutions and observed that the former DES dissolved more lignin than the latter one regardless of the used HBDs, indicating the stronger lignin dissolving capability of C = C functional group than OH group contained in HBA. The observations of Liu et al. (2019) seemed to be inconsistent with our findings that DES with HBA containing OH group removed more lignin than that containing C = C group. This indicates that a solvent that can dissolve more extracted lignin does not mean to demonstrate better pretreatment performance. Our assumption was confirmed by Kwok et al. (2019), who reported that there was no correlation between lignin solubility in organic solvents and delignification efficacy during pretreatment with those solvents. The plausible reason for this phenomenon was that the structure of lignin separated from cellulosic material was different from that embedded in raw biomass, because the latter lignin was covalently cross-linked with carbohydrate (especially hemicellulose) fraction through benzyl ester, benzyl ether and phenyl glycoside functional groups (Balakshin et al., 2014).

3.3. Enzymatic hydrolysis of the pretreated biomass

After pretreating CS with LA-based DES composed of differently structured HBA, the obtained solid residues were saccharified enzymatically to evaluate the effect of different pretreatments on enzymatic cellulose hydrolysis and the results are shown in Fig. 1. As expected, the unpretreated CS demonstrated the lowest glucose yield throughout the entire time-course of enzymatic hydrolysis, indicating the highly recalcitrant structure of raw CS sample. By contrast, after DES pretreatment, enzymatic cellulose conversion was greatly improved, and the enhancement degree was highly dependent on alkyl chain length of HBA used for preparation of DES. The longer the alkyl chain length of HBA, the lower the glucose yield. TMAC (short alkyl chain)-LA DES pretreatment afforded a cellulose-to-glucose conversion of 73.30% after 48 h enzymatic reaction, whereas at the same hydrolysis time, cellulose saccharification rate decreased to 50.89% when using TBAC (long alkyl chain)-LA pretreated CS for hydrolysis (Fig. 1a). The same conclusion could also be obtained by comparing the enzymatic hydrolysis of CS pretreated with DES containing HBA with benzyne ring. BTMAC has shorter alkyl chain than BTEAC, thus pretreatment with DES consisting of BTMAC demonstrated a higher glucose yield than its counterpart containing BTEAC (72.60% vs 66.67%, Fig. 1b). The functional group of HBA also exhibited dramatic effect on enzymatic digestibility of corresponding pretreated CS samples. As shown in Fig. 1b, a significantly higher glucose yield (73.42%) was attained from enzymatic hydrolysis of ChCl-LA pretreated CS compared to ATMAC-LA and CCC-LA pretreatments, which gave rise to 62.16% and 65.53% enzymatic cellulose conversion rate of the pretreated CS at the same hydrolysis time of 48 h. This corroborated that the pretreatment with DES containing hydroxyl HBA was able to produce more enzymatically digestible biomass substrate than DES with HBA containing chloro and carbon–carbon double bond groups.

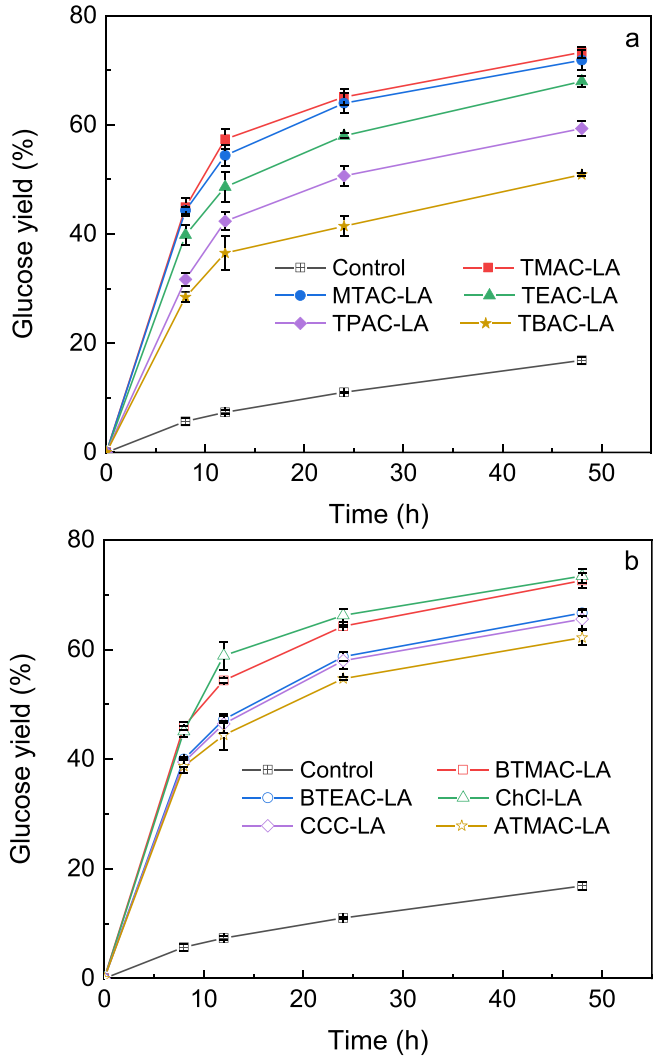


Fig. 1. Enzymatic hydrolysis of pretreated biomass with different DESs.

CS pretreatments with DESs composed of differently structured HBA led to delignification and xylan removal at varied degrees, thus it was intriguing to investigate the correlations between removal of xylan and lignin and enzymatic saccharification of corresponding pretreated solids. It was observed that both delignification and xylan removal were shown to have relatively strong and positive relationships with enzymatic cellulose hydrolysis as indicated by the coefficient of determination (R^2) of higher than 0.81 for both cases. Comparatively, the R^2 value between delignification and enzymatic cellulose hydrolysis was higher than that between xylan removal and enzymatic cellulose hydrolysis (0.8764 vs 0.8161), implying that lignin removal demonstrated a more profound effect on enzymatic cellulose digestion than xylan removal. Those observed phenomena were expected by taking into the consideration of the fact that both xylan and lignin serve as physical barriers preventing the contact of cellulolytic enzymes with cellulose fraction. With large amount of xylan and lignin being removed after DES pretreatments, the cellulose accessibility was markedly enhanced, thus leading to great improvement in subsequent enzymatic digestibility of pretreated biomass (Zhao et al., 2012). On the other hand, apart from acting a physical barrier, lignin remaining in pretreated solids can also non-productively adsorb cellulase during enzymatic hydrolysis process. As a result, more delignification during pretreatment could enable substantial reduction in non-productive adsorption of cellulase on the residual lignin, allowing more amount of free enzyme to hydrolyze cellulose fraction present in pretreated biomass (Li et al., 2016; Yoo et al., 2020).

3.4. Correlations between DES properties and pretreatment performance

In theory, the varied pretreatment performance of different DESs must be linked to their different intrinsic physical and chemical properties. The commonly used physical parameters to describe DES solvents include density, viscosity, conductivity and pH. The chemical properties of DES solutions can be well described by three Kamlet-Taft solvatochromic parameters, i.e., α, β and π*. In the present work, physicochemical properties of the ten tested DESs were correlated to their pretreatment performance in terms of xylan removal, delignification and cellulose digestibility.

Density values of the DESs was found to have a good linear relationship with xylan removal during pretreatment with R^2 value equal to 0.71 (Fig. 2a) but showed weak linear correlations with delignification ($R^2 = 0.35$) and cellulose saccharification ($R^2 = 0.51$). Available reports on the effect of density of a solvent on its penetration into biomass matrix were conflicting. Jablonsky et al. (2019) pointed out effective penetration was attained from a solvent with low density. On the contrary, Zhang et al. (2019) believed that high density of a solvent was conductive to penetration. In our study, high xylan removal was attained from biomass pretreatment by DESs with high densities, which was in good agreement with the report of Zhang et al. (2019). Viscosity values of DES was observed to have no clear or low correlation with xylan removal, delignification and enzymatic cellulose hydrolysis, and so was the case for values of conductivity and pH of the tested DES solutions. It was worth pointing out that all above mentioned physical properties of the DESs were measured at room temperature. In the present work, DES pretreatment of biomass was carried out at 130 °C. At this high temperature, the property values might be significantly different from that determined at room temperature. Take viscosity for example, it is an established fact that viscosity of a solvent decreases with increasing temperature. The large difference in the viscosity of DES measured at room temperature would be significantly narrowed at the high temperature of 130 °C used for CS pretreatment. This would explain the no or low relationship between DES viscosity and biomass pretreatment performance observed in the present work.

It is widely accepted that the hydrogen bond basicity (Kamlet-Taft parameter β) of ionic liquids has a significant effect on the dissolution of lignocellulosic biomass (Brandt et al., 2013; Doherty et al., 2010). Regarding the relationship between lignin extraction/dissolution and Kamlet-Taft parameters of DESs, different conclusions were obtained from different reports using different substrates as mentioned in the Introduction section. In the present study, Kamlet-Taft parameter π* values of the DESs were not relevant to their pretreatment performance. Parameter β values of DESs were found to be poorly correlated to removal of xylan and lignin and cellulose saccharification with R^2 values less than 0.45 for those cases. By contrast, strong positive linear relationship was observed between Kamlet-Taft parameter α values of DESs and their pretreatment performance as implied by the high R^2 values of established equations which were calculated to be 0.73, 0.75 and 0.89 for xylan removal, delignification and enzymatic cellulose conversion, respectively (Fig. 2b-2d). Among the three Kamlet-Taft parameters of DES solutions, α values were observed to have the most significant effect on pretreatment efficiency of lignocellulosic biomass. The reason for this observed phenomenon could be explained by the great contribution of acidity of DESs to deconstruct lignocellulosic structure. It was reported that biomass pretreatment mechanism of an acidic DES involves proton-catalyzed cleavage of ether, ester and glycosidic linkages present in lignin-carbohydrate complex, leading to extraction and depolymerization of lignin and xylan (Satlewal et al., 2018; Xu et al., 2020). The proton catalyst is stemmed from dissociation of LA, which acts as HBD of

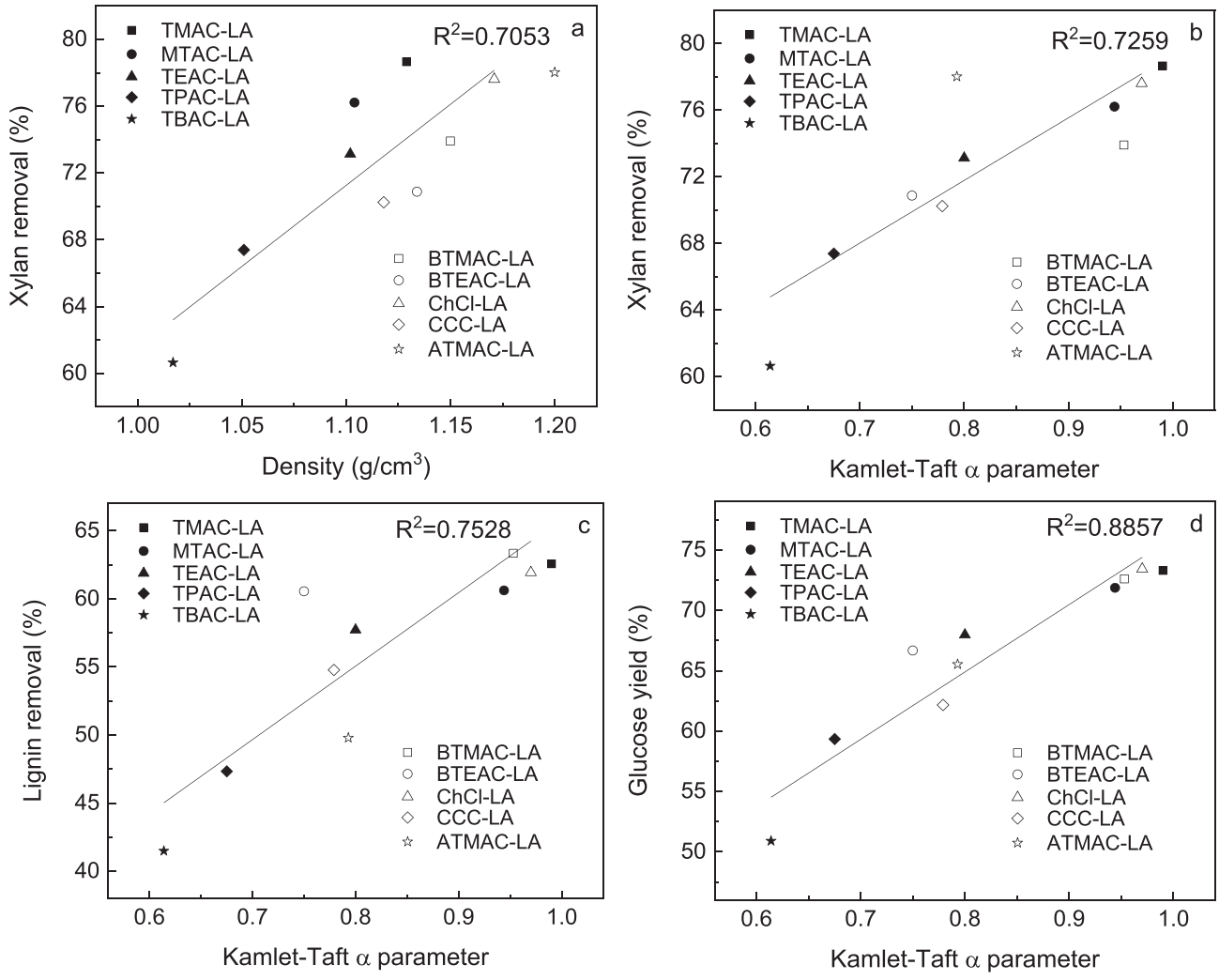


Fig. 2. Correlation between density and xylan removal (a) and correlations between Kamlet-Taft α parameter and removal of xylan (b), delignification (c) and enzymatic hydrolysis of pretreated solids (d).

DES. Although all the tested DES contained the same amount of LA and chloride ions (contained in HBAs), pretreatments with DESs synthesized from LA and HBAs with different quaternary ammonium cations resulted in different levels of removal of xylan and lignin and cellulose conversion. These results suggested the important roles that quaternary ammonium cations of HBA play in pretreatment efficacy of DES solutions. Smink et al. (2019) found that ChCl-LA pretreatment of *Eucalyptus globulus* chips removed more lignin than single LA pretreatment. They concluded that chloride anion present in ChCl of DES, contributed a lot to delignification process. The important role that halide anions of HBAs played in lignin extraction during DES pretreatment was elucidated in detail by da Costa Lopes et al. (2020). They reported that nucleophile substitution of hydroxyl groups of 2-phenoxy-1-phenylethanol (PPE), a lignin model compound, by halide anion allowed to form halide-PPE intermediates, which was ready for cleavage of β-O-4 linkage in model compound in the presence of proton catalyst. Our study further found that quaternary ammonium cations of HBAs, in addition to halide anions of HBAs, also had determinant effect on their biomass pretreatment performance. As can be seen from the high correlation between Kamlet-Taft α parameter and removal of lignin and xylan, it could be deduced that the quaternary ammonium cations of HBAs might affect the proton-donating ability of the DESs. High proton-donating ability, i.e., high Kamlet-Taft parameter α values, is favorable for proton release to cleave chemical bonds in lignin-carbohydrate complex. Also, according to the reports of da Costa Lopes et al. (2020), the quaternary ammonium cations of HBAs might have some influence on the formation of lignin-chloride intermediate, thus in turn affecting the proton-catalyzed depolymerization of lignin in the present study.

3.5. Sugars and inhibitors in the pretreatment liquids

During lignocellulose pretreatment with thermochemical methods under acidic conditions, the fractions of cellulose and hemicellulose can be partially degraded to their constituent monosaccharides, i.e., glucose and xylose. As shown in Table 3, in addition to pretreatments with BTMAC-LA, BTEAC-LA and CCC-LA (for those three pretreatments, glucose contents could not be accurately determined by HPLC due to impurity intervention), all the other DES pretreatments resulted in pretreatment streams with considerably higher xylose contents than glucose contents, suggesting that hemicellulose is more liable to proton-catalyzed degradation than cellulose. Acetic acid present in the pretreatment hydrolysate is formed from hydrolysis of acetyl group of hemicellulose. Its concentrations were higher than 1 wt% in the liquids of pretreatments with TEAC-LA, TPAC-LA, TBAC-LA, BTEAC-LA and ChCl-LA. Furthermore, the xylose and glucose released from depolymerization of polysaccharide fraction could be further dehydrated to form furfural and 5-hydroxymethylfurfural (HMF), respectively. As can be seen from Table 3, HMF concentrations varied in a narrow range between 0.49 and 0.69 wt%, whereas furfural contents varied a bit largely from 0.44 to 1.58 wt%. The concentrations of the three commonly found inhibitors in pretreatment liquids were significantly lower than other commonly used chemical methods when pretreating corn stover, such as dilute acid (Hodge et al., 2008), hydrothermal (Xu et al., 2010) and wet explosion (Rana et al., 2014).

Table 3
Contents (wt%) of monosaccharides and inhibitors based on dry weight of substrate in different pretreatment liquids.

No.	DES	Glucose	xylose	Acetic acid	HMF	Furfural
1	TMAC-LA	3.94	6.98	0.63	0.62	1.06
2	MTAC-LA	1.74	3.41	0.54	0.55	0.95
3	TEAC-LA	2.93	5.71	2.53	0.61	0.83
4	TPAC-LA	0.85	1.75	2.08	0.62	0.61
5	TBAC-LA	0.55	1.14	1.82	0.59	0.44
6	BTMAC-LA	NP	5.39	0.41	0.63	0.88
7	BTEAC-LA	NP	4.21	2.17	0.62	0.63
8	ChCl-LA	2.77	4.80	3.10	0.57	0.79
9	CCC-LA	NP	7.13	0.68	0.69	1.58
10	ATMAC-LA	2.70	5.18	0.52	0.49	1.08

NP stands for that glucose concentration could not be obtained due to irregular HPLC peak shapes.

3.6. Yields and associated polysaccharides contents of DES lignin samples

During pretreatment of CS, a certain amount of lignin in raw biomass could be extracted by DES into the pretreatment liquid. This part of lignin was collected and referred to DES lignin. In order to make the overall lignocellulosic biorefinery process economically viable, the lignin fraction, in addition to polysaccharide fraction, should also be efficiently recovered and fully utilized. Table 4 shows the yields and associated polysaccharide contents of the DES lignin samples. It was obvious that no matter the quaternary ammonium cations contain benzene ring or not, the DES lignin yields declined with increasing alkyl chain length. With regard to pretreatment with DES containing HBA with different functional groups, the DES lignin yields decreased in the following order according to functional groups: –OH > -C = C > -Cl. All these results were in good agreement with delignification data reported in Table 2. The DES lignin yields from CS pretreatments with the ten types of DES varied between 59.53% and 81.47%. BTMAC-LA and ChCl-LA pretreatments resulted in DES lignin with yields higher than 80%. The lignin yields reported in the present study were significantly higher than hydrothermal pretreatment (Xu et al., 2018), ethanol organosolv pretreatment (Bauer et al., 2012) and some ionic liquid pretreatments using commercial 1-butyl-3-methylimidazolium chloride or 1-butyl-3-methylimidazolium methylsulfonate with hydrochloric acid as catalyst (Moghaddam et al., 2014) and choline acetate (Ninomiya et al., 2015). Lignin yields in all these reports were found to be less than 50%. Our result was in good accordance with previous reports by Chen and Wan (2018) and Cronin et al. (2020), who obtained about 71% and 75% lignin yields resulting from DES pretreatments of CS, respectively. The purity of lignin is a crucial issue for its value-added utilization. The main polysaccharide contents associated with DES lignin were detected and the results are shown in Table 4. In addition to CCC pretreatment, xylan with contents ranging from 0.81% to 1.53% was found to be the main polysaccharide associated with lignin in all the DES lignin samples, whereas glucan was present in small amounts. This was expected given the fact that in raw biomass, native lignin is covalently linked to hemicellulose, conferring mechanical strength to the plant cell wall (Terrett & Dupree, 2019). As aforementioned, DESs mainly attack the ether and ester bond contained in lignin-hemicellulose complex, releasing DES lignin containing residual xylan that is not completely depolymerized into pretreatment liquids. The polysaccharide contents associated with DES lignin samples coincided with the results of Guo et al. (2019), who reported the xylose contents of lignin fractions ranged from 1.5% to 1.98% under all the tested conditions. The relatively low polysaccharide contents associated with DES lignin indirectly confirmed

Table 4
The yields and associated polysaccharides contents of DES lignin samples.

No.	Pretreatment	Yield (%)	Associated polysaccharides (wt%)	
			Glucan	Xylan
1	TMAC-LA	77.72	0.06	1.02
2	MTAC-LA	76.01	0.28	1.33
3	TEAC-LA	68.35	0.16	1.53
4	TPAC-LA	64.18	0.23	1.34
5	TBAC-LA	59.53	0.17	1.22
6	BTMAC-LA	81.47	0.01	0.81
7	BTEAC-LA	79.84	0.10	1.10
8	ChCl-LA	80.32	0.02	0.93
9	CCC-LA	72.39	0.10	ND
10	ATMAC-LA	77.53	0.11	1.05

ND stands for not detected.

their high purity. Plus taking into account of the high extraction yield of DES lignin, it can be believed that the extracted DES lignin in the present study would be excellent raw feedstock for valorization to different products via chemical and/or biochemical routes.

4. Conclusions

Ten types of LA-based DESs containing HBAs with different structures were used to pretreat CS. Results showed that both alkyl chain length and functional groups of HBAs significantly affect pretreatment effectiveness of DES. The chemical properties of the DESs were characterized by Kamlet-Taft solvatochromic parameters. Correlation study on the property-performance of the DESs showed that hydrogen bond basicity (parameter α) had a positive linear relationship with pretreatment efficacy of the DESs. The clarification of structure–property-performance relationships of the DESs would benefit efficient design and selection of DESs for lignocellulose pretreatment.

CRediT authorship contribution statement

Xinquan Liang: Investigation, Methodology, Data curation, Validation, Supervision. **Yuan Zhu:** Investigation, Validation, Visualization. **Benkun Qi:** Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing, Supervision. **Shiqian Li:** Data curation, Visualization. **Jianquan Luo:** Writing - review & editing. **Yinhua Wan:** Resources, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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